

Amplify AR HUD User-Experience with Real-World Sunlight Simulation in Virtual Scene

Eun Hye Kim*
EC Lab, Hyundai Mobis, Republic of
Korea
eunhye.kim@mobis.com

Hyocheol Ro
EC Lab, Hyundai Mobis, Republic of
Korea
hyocheol.ro@mobis.com

Hyunjin Park
EC Lab, Hyundai Mobis, Republic of
Korea
hjpark@mobis.com

ABSTRACT

In this paper, we introduce the concept of real-world sunlight simulation in virtual scenes of Head-Up Display (HUD) application. We have adopted solar position calculation logic [Zhang, 2020] and demonstrated its capability with experiments ran on real vehicle (Figure 1). In addition, we suggest UX approaches featuring the interaction between virtual 3D objects and the simulated virtual sunlight for expanded AR experiences in HUD application.

CCS CONCEPTS

• **Human computer interaction (HCI);** • **Interaction paradigms;** • **Mixed / augmented reality;**

KEYWORDS

Head-Up Display, In-Vehicle Infotainment

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1 MOTIVATION

Among the commonly recognized building blocks to form a virtual environment, the most fundamental element is lighting. Commercial AR toolkits, such as ARCore [Google, 2020], provide and highlight the lighting APIs for integrating physical world's light into virtual scenes.

We had embarked on AR HUD project on 2022 and the goal was to explore AR application design methods applicable on real vehicle HUD. In the first step, we referenced popular mobile AR applications and tried to visualize similar lighting effects: shadow, surface reflection, and shading. However, we soon recognized it was a wrong approach just to mimic the visual appearance from applications which are originally designed to run on structurally different devices.

Only after clarifying the distinctive HUD lighting condition determined by its hardware structure and use cases, we could derive

one basic principle for AR HUD lighting: the source of light in virtual scene should be mapped with physical world's sunlight real time. We provide detailed explanation with comparison to smartphones in Table 1.

We are expecting our work would help the future developers, who would newly experience HUD, get coherent understanding of HUD lighting features prior to designing and development. This will help them avoid mistakes that easily happen when focusing only on the term 'AR' and ignorant of basic concepts of HUD.

2 RELATED WORK

AR application had rapidly gained popularity and diverged into many forms of contents running on different devices. Especially, smartphones domain has established matured echo system with active contribution both from API providers and application developers. We described the lighting related condition in smart phone devices in Table 1 to put emphasis on distinctive characteristics that HUD has as an outdoor transparent display.

3 PROBLEM DEFINITION

The key part in AR application is to ensure virtual objects are under the same condition of real-world lights. However, the source of light in physical world is now a single solar while conventional AR UX design techniques for realistic lighting are based on multiple, room-scale indoor lights. We could not adopt the concept of light estimation techniques from smartphones domain into outdoor AR HUD. Therefore, new methods were required to align virtual object's shadow casting and illumination with physical surroundings.

4 SOLUTION

We combined Solar Formula and real time input from car (Algorithm 1). Above logic is demonstrated and implemented on real vehicle HUD (Figure 1).

5 CONTRIBUTION AND EXPECTATION

Since vehicle as a development environment is not easily accessible for application developers, our work would help future AR HUD developers by leading them not to repeat the trial and error we already had gone through. In addition, solar formula-based approach is expected to assist computer vision-based inverse light methods with lower computing costs.

6 LIMITATION AND FUTURE WORK

When it is after sunset, the real world's light source becomes more complicated. Multiple artificial lights are generated by streetlamp, surrounding cars, buildings, and etc. Advanced sensors and fusion algorithm should be explored. Additionally, in the shadowed area

*Corresponding Author

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Table 1: AR Application Device Comparison

	Smart Phone	Head Up Display
Scale	Room scale. [Google, 2020]	Anywhere accessible by car.
Source of Light	In-door lights.	Sun.
Display	Virtual scene mapped on real world image captured by camera.	Transparent. Augmented content visualized as overlay on windshield front glass [Blankenbach, 2023].
Light Reflection	The transform of virtual light is estimated based on camera input image.	Not actively applied in commercial AR HUD application.



Figure 1: (a) is recorded at 11:00 am. (b) is recorded at 16:00 pm in the same place. The virtual objects cast shadow in the direction aligned with surrounding physical objects. The shadow casted by parked cars shows different angle on the road surface, which indicates the solar position has changed over time. The sphere has metallic surface so that it reflects simulated sun in virtual skybox. Please refer to the supplementary video recorded while real time driving.

Algorithm 1 Calculate the Direction of Sunlight in Virtual Scene

Input

Car heading angle (following the South-Clockwise convention),
current vehicle position in latitude and longitude (ϕ_o, λ_o).

Output

Euler angle for lighting system in game engine (Unity's
Directional Light component which casts light toward z axis
positive direction).

Coordinate System

Cartesian coordinate system. The origin (0,0,0) is the center of
vehicle.

Steps

- 1) Get current car heading angle, latitude, and longitude from dedicated GPS sensor.
- 2) Calculate Solar Elevation and Azimuth from formula [Zhang, 2020].

Subsolar Point's Coordinates

2-1) ϕ_s, λ_s is the declination of the sun.

2-2) $\lambda_s = -15(TGMT - 12 + E \text{ min}/60)$

2-3) $Sx = \cos \phi_s \cdot \sin(\lambda_s - \lambda_o)$

2-4) $Sy = \cos \phi_o \cdot \sin \phi_s - \sin \phi_o \cdot \cos \phi_s \cdot \cos(\lambda_s - \lambda_o)$

Solar Zenith :

$$S_z = \sin \phi_o \sin \phi_s + \cos \phi_o \cos \phi_s \cos(\lambda_s - \lambda_o)$$

$$Z = \cos^{(-1)} S_z$$

Solar Azimuth angle following the South-Clockwise convention:

$$\gamma_s = \text{atan2}(-Sx, -Sy)$$

- 3) Coordinate Conversion (Calculated data from 2 is based on different origin):

$$\gamma_s' = (\gamma_s - \text{Car Heading Angle})$$

- 4) Unity Directional Light 's Euler Angle = new Vector3 (90-Z, γ_s' , 0)

where the real-world sunlight is blocked by surroundings, virtual objects should be shaded. It requires shadow recognition logic with low latency. We have yet to explore more in image recognition algorithms.

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